Abstract

Many lightweight tactical UAVs have difficulty finding a compromise between directional stability and drag, while operating at Low Reynolds numbers close to Re ≈ 66,000. This can lead to flow separation and the formation of laminar separation bubbles, which can reduce aerodynamic performance. To address this problem, a computational framework was developed to investigate how different vertical tailfin geometries affect aerodynamic performance. The framework combines parametric geometry generation using CadQuery, Reynolds-averaged Navier-Stokes (RANS) simulations in OpenFOAM using the k-ω SST-LM model, Gaussian Process Regression (GPR) surrogate modeling, and NSGA-II multi-objective optimization. Near-wall mesh resolution was maintained with a mean y+ value of 1.6 (localized peaks less than 59) as described in Section 2.2. The surrogate model achieved an R^2 value of 0.991. The near-optimal design, run_25, achieved a 59% increase in yaw-moment derivative and a 39% increase in the side-force derivative, alongside a 24% increase in drag. This result was then compared to the standard RQ-11 Raven vertical tailfin, with the same freestream velocity of 10 m/s. These results correspond to the yawing moment coefficient derivative (Cnβ), side-force coefficient derivative (CYβ), and drag coefficient (Cd), respectively. This resulted in a 28% increase in restoring force per unit drag. Mesh independence was evaluated using the Grid Convergence Index (GCI), with the fine-grid result producing a GCI of 4.9%. Additional validation points were also tested against full RANS simulations, with the surrogate model producing errors of +1.0% and −0.3%. Flow-field analysis showed that the boundary layer remained attached at zero incidence.

Introduction

2. Methodology & Computational Pipeline

2.1 Parametric Geometry Generation (CadQuery)

All candidate vertical tailfin geometries were generated programmatically using a custom CadQuery parametric script. The geometric design space is governed by three primary non-dimensional shape parameters: quarter-chord sweep angle (Λ), aspect ratio (AR) and taper ratio (λ).

To maintain a consistent physical scale across the design space, the root chord (c_{root}) was fixed at 0.15 m. Tip chord and total span (b) were derived analytically according to:

$$C_{tip} = C_{root} \cdot \lambda$$

$$b = AR \cdot c_{mean}$$

C_{mean} represents the arithmetic mean chord. Spanwise sweep was implemented as a linear spatial offset applied to the tip profile. The resulting 3D volume was created as a straight ruled surface loft between the root and tip sections without spanwise twist or relative thickness variations.

Every candidate geometry in the optimization family utilizes the symmetric SD8020 airfoil profile, at around 7% thickness. To prevent high-aspect-ratio cell distortion and numerical divergence along the aft section during volume meshing, the sharp trailing edge from the raw coordinate file was blunted to a uniform 0.1% chord gap fraction across both root and tip sections.

2.1.2 Benchmark Comparison Framing

To benchmark performance against a representative tactical low-Re airframe, an RQ-11 Raven vertical tailfin geometry was constructed with the same CAD pipeline. The benchmark retains the stock Raven physical envelope: 100mm root chord, 60mm tip chord, 150 mm span, taper ratio of 0.6, sweep angle of 15°, and a wetted planform area of 0.0120 m^2, utilizing an analytic NACA 0009 airfoil with 9% relative thickness.

By contrast, the near-optimal reference design, run_25 (case_25_b+000, Cd = 0.01085), serves as the non-extrapolated reference anchor; the surrogate-predicted optimum case_99 (Cd = 0.00998) lies outside the training hull and is flagged as extrapolated. run_25 features an SD8020 profile at 7% thickness and a larger planform area of 0.0174 m^2, versus the Raven's NACA 0009 at 9% thickness and 0.0120 m^2. Methodological fairness between the two distinct geometries is maintained by evaluating both fins under identical numerical conditions: matched freestream velocity ($U\infty$ = 10 m/s, yielding Re ≈ 66.6k for run_25 and Re ≈ 44k for the Raven), identical snappyHexMesh boundary layer refinement rules, identical force reference parameters (A_{ref}, l_{ref}), and identical solver convergence criteria. Comparative performance and efficiency claims are evaluated strictly through per-drag force normalization ($C_{n\beta}/C_d$ and $C_{Y\beta}/C_d$).

2.2 CFD Solver Setup & Mesh Configuration

Numerical simulations were conducted using OpenFOAM v2606 running simpleFoam, a steady-state incompressible solver implementing the SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm. Turbulence was modeled using the four-equation k-ω SST-LM transition model. To make sure the solutions were settled, a dual-gate convergence check was used: residuals had to drop below 10^{-4} across all fields after running for at least 300 steps, and the force coefficients (C_Y, C_d, C_n) had to completely flatten out.

The background mesh box was set from (-0.5, -0.5, -0.1) m to (1.5, 0.5, 0.5) m, making a total volume of 2.0 by 1.0 by 0.6 m. Compared to the 0.15 m root chord, this put the inlet about 3.3c

in front of the fin and the outlet about 9c behind it in the wake. These dimensions were kept compact to save compute time while avoiding blockage issues.

For boundary conditions, the inlet used a fixed velocity, the outlet was set to a fixed-value kinematic pressure outlet ($p = 0\ m^2/s^2$) , and the sides and top used slip conditions. The bottom plane at z = -0.1m was set as a symmetry plane, leaving a 0.1m gap between it and the tailfin root at z = 0 so it didn't create a fake wall junction. Sideslip angles were tested by rotating the incoming velocity vector. Everything was normalized using a reference area of 0.05 m^2 and a reference length of 0.149927m, with moments measured from the quarter-chord root.

The meshes were built starting from a blockMesh background grid (80 x 40 x 20, or 64k base cells) and refined using snappyHexMesh. This produced final cell counts between 200k and 420k for the design variations (for instance, the optimal design had 205,802 cells), with a hard cap set at 600k.

Viscous boundary layers were resolved using 12 inflated prism layers growing at a 1.2 expansion ratio from an initial first-cell height of 15μm, achieving around a 98% layer coverage. Regional wall-normal resolution yielded local y^+ means ranging between 0.7 and 3.1, with stagnation and high curvature ridge zones reaching peaks of 30-70. In all, it maintained high fidelity across attached boundary layers.

2.3 Design of Experiments and Data Collection

The design space was sampled with a 30-point Latin Hypercube over sweep, between 15° and 35°, aspect ratio between 1.5 and 3.0, and taper between 0.4 and 0.8. Each row was built into a complete case directory from the frozen template by an orchestration script, which consisted of fresh CAD export at coarse tesselation, trimesh watertight repair, blockMesh, surfaceFeatureExtract, and snappyHexMesh -overwrite, followed by simpleFoam under the monitored convergence policy (300-iteration floor, 1000-iteration harbor, stopping only when all field residuals read below 10^{-4} across consecutive checks and drag and lift flatten over a 50-iteration window).

Every case ran inside exception handling, so any meshing failure or floating-point divergence was logged to an error file, marked FAILED in the master table, and the sweep continued without intervention. Force coefficients were logged every iteration through two function objects: a longitudinal forceCoeffs and a lateral forceCoeffsLateral whose lift and pitch channels read out side force (C_Y) and yaw moment (C_n) about the quarter-chord root pivot.

Flow regime checks ran as post-processing: per-face y^+ mapped to local chord fraction with statistics over the 30-70% chord band, plus a skin-friction sign analysis for separation lines (with

the sign convention noted: OpenFOAM reports wall shear on the fluid, so attached flow reads negative and reversed bubble flow reads positive). The final dataset holds 30 converged cruise rows, 18 sideslip rows across two geometries, and 20 high-sweep delta rows, each reproducible from its committed system dictionaries and STL files.

2.4 Surrogate Modeling and Optimization Setup

Four response targets (C_d, C_l, LSB-zone y^+ mean, y^+ average) were each fitted with three model families: Gaussian Process Regression with a Matérn kernel, gradient boosting, and quadratic polynomial response surfaces with ridge regularization. Models were compared by 5-fold cross-validated R^2 with out-of-fold RMSE and MAPE plus leave-one-out RMSE, and the winning model per target was retained (GPR for drag at $R^2 = 0.991$, gradient boosting for the weak bubble proxy at 0.69). The global optimum came from dense random search over the joint proxy,

$$J' = C_d + \alpha \cdot LSB_{mean}$$

where α is the bubble weight, noting in the run logs that axial-flow data contains no true side force or bubble-length terms, so J' stands in for the proposal's full objective until the sideslip campaign supplies them.

Multi-objective tradeoffs were traced with NSGA-II (population size 100, 100 generations) over the proposal bounds, with every Pareto point outside the training hull explicitly flagged as extrapolated rather than trusted. Global sensitivities came from Saltelli sampling (N = 1024, second order, 8192 surrogate evaluations per target): aspect ratio carries 64% of drag variance and taper carries 96% of the bubble proxy, with total-effect indices matching first-order ones, indicating additive parameter behavior. Interior validation points, never seen in training, matched full RANS within 1.0% and -0.3%, and a rerun determinism check reproduced drag to five digits.

2.5 Verification of Mesh, Model and Repeatability

Grid independence was established on the optimum geometry at three refinement levels (85k/206k/448k cells, constant 15μm first layer): $C_d = 0.01128/0.00998/0.00951$, monotonic with a fine-grid GCI of 4.9% and a Richardson-extrapolated value of 0.00913. Turbulence-model sensitivity analysis shifts drag by 2.1-2.5% while preserving design rankings. Computational determinism was verified by rerunning cases to identical drag values across five digits. A known precision floor arises from mesh asymmetry: zero-sideslip reference rows carry residual side forces of approximately 3×10^{-3}, bounding single-degree derivative claims at roughly 5%.

3. Results

3.1 Pareto Frontier and Multi-Objective Design Space Topography

3.1.1 Macro-Level Trade-Offs and Degenerate Corner Collapse

The NSGA-II multi-objective optimization mapped the non-linear tradeoffs between parasitic drag (C_d) and boundary layer separation severity, which was tracked via the LSB-zone proxy. As illustrated by the surrogate-driven Pareto analysis, the feasible design space exhibits a distinct degenerate corner collapse. Within the parameter space, taper ratio exhibits a strong positive correlation with the bubble proxy: lower taper ratios shrink tip chords and suppress the LSB proxy, whereas less tapered planforms broaden the outbound chord and expand the mid-span separation footprint. Aggressively reducing tip chord length lowers the local tip-region wetted area, reducing the spatial extent over which an LSB can form, but forces a steeper local adverse pressure gradient that trades bubbles for localized profile drag penalties.

The resulting Pareto frontier isolates three critical design markers: the absolute minimum-drag floor, a balanced knee point governing optimal stability-to-drag scaling, and the minimum LSB boundary. Because extracting directional derivatives required multi-angle velocity vector rotations across target sideslip cases rather than the full 30-point DoE matrix, the surrogate proxy space was used to select candidate geometries for detailed directional evaluation. Within this proxy space, the near-optimal candidate case_99 achieves a measured confirmation drag of C_d = 0.00998. For raw DoE verification, run_25, with a C_d of 0.01085 serves as the non-extrapolated reference anchor.

3.2 Aerodynamic Performance vs. RQ-11 Raven Baseline

To quantify the performance advantages of non-linear planform shaping, the near-optimal reference geometry case_25 was evaluated directly against the RQ-11 Raven baseline under identical freestream conditions, stated before as Re = 66.6k, with U_∞ = 10 m/s. Crucially, moments for both geometries are evaluated about their respective quarter-chord root pivots, evaluating the Raven about its historical root-chord pivot point (37.5%) artificially yields a destabilizing negative $C_{n\beta}$ slope, whereas evaluating about its true quarter-chord establishes a valid baseline.

This enhanced suction distribution stems from two primary structural factors: transitioning from the stock 9% thick NACA 0009 to the 7% thick SD8020 profile reduces initial frontal blockage, while non-linear spanwise redistribution shifts wetted planform area farther outboard. When

subjected to sideslip, this additional outboard area acts across a longer physical moment arm relative to the quarter-chord root pivot. This spatial redistribution directly explains the disproportionate force gains, driving a 59% increase in yaw stiffness and a 39% increase in side-force derivative.

Evaluating airframe performance strictly through raw parasitic drag obscures directional control authority, as run_25 carries a larger wetted envelope of 0.0174 m2 compared to the baseline 0.0120 m2. These directional improvements incur a 24% drag tax in absolute parasitic drag (Cd = 0.01085 vs. Raven Cd = 0.00872). However, normalizing performance through per-drag efficiency metrics yields a net 28% efficiency gain in Cnβ/Cd and a 12% gain in CYβ/Cd over the legacy RQ-11 tailfin (±5% numerical bound). This confirms that strategic planform shaping breaks traditional linear trade-off curves, delivering superior directional stability without incurring a proportional drag penalty.

3.3 Flow-Field Physics and Boundary Layer Behavior

Within the operational regime of Re ≈ 66.6k, viscous forces predominate relative to fluid inertia, rendering conventional planforms highly susceptible to extensive laminar separation bubbles (LSBs) under adverse pressure gradients. Nevertheless, spatial flow-field post-processing demonstrates that boundary layer flow across the optimized tailfin series remains entirely attached at zero incidence. Chordwise pressure maps show that short-intense planforms (run_25 and Raven) exhibit sharp, short pressure recoveries, whereas the global optimum exhibits a longer, flat pressure plateau with values of 0.38c and 0.83c, respectively. This contrast confirms that low-Re directional stability gains are governed primarily by planform area redistribution and skin-friction management rather than active LSB suppression dynamics. Evaluation of wall shear stress (τ_w) across the critical 30–70% chordwise region corroborates this fully attached state. Under OpenFOAM, wall shear stress is evaluated on the fluid, where negative values indicate attached flow and positive values indicate reversed flow. Across run_25, skin-friction diagnostics show persistently negative values, confirming that no flow separation occurs at cruise. Off-body flow-field evaluations sampled across two constant-x crossflow cutting planes and reduced to maximum Q-criterion statistics show no high-intensity concentrated vortex cores near the tip. All optimization evaluations were conducted at zero angle of attack. While sideslip polars quantify directional restoring moments, complete angle-of-attack performance polars were omitted due to geometric top-bottom symmetry at cruise.

3.4 Surrogate Model Fidelity and Global Parameter Sensitivities

Gaussian Process Regression (GPR) with a Matern kernel yielded high predictive accuracy for integrated aerodynamic force coefficients: $R^2 = 0.991$ for drag (C_d) and $R^2 = 0.998$ for axial lift (C_l). For the reason that axial lift on a symmetric vertical tailfin at zero incidence operates on a

near-zero noise flor at around 10^{-4}, high R^2 values reflect strict numerical symmetry in the solver rather than structural predictive complexity. In contrast, modeling the localized boundary layer separation proxy via gradient boosting yielded a reduced R^2 of 0.69, underscoring the heightened numerical sensitivity of RANS-based transition transport equations in resolving discrete boundary layer physics. Global variance-based sensitivity decomposition using Saltelli sampling (N = 1024, corresponding to 8,192 surrogate evaluations) established that aspect ratio accounts for 64% of total drag variance, whereas taper ratio governs 96% of boundary layer separation characteristics. Total-effect sensitivity metrics closely mirrored first-order indices, demonstrating minimal multi-variable non-linear coupling. Consequently, geometric inputs exhibit largely additive behavior, establishing a highly predictable and robust topology suitable for real-world manufacturing constraints in low-Reynolds-number tailfin design.

4. Limitations and Future Work

4.1 Limitations of the Computational Framework

4.1.1 Transition Model Calibration Limits at Re ≈ 66.6k

While utilizing the four-equation k-ω SST-LM model provides essential resolution for laminar-to-turbulent transition, its underlying empirical correlations are calibrated primarily on higher-Reynolds-number datasets. At Re ≈ 66.6k, some sensitivities in predicting the critical momentum-thickness Reynolds number can shift predicted separation and reattachment locations. Consequently, the LSB proxy serves as an effective relative optimization guide across the design space, but absolute transition locations carry inherent numerical uncertainty.

4.1.2 Surrogate Proxy vs Direct Derivative Optimization

Computing full sideslip polars across the entire 30-point Design of Experiments matrix was computationally prohibitive. Consequently, the optimization loop relied on an auxiliary surrogate proxy. with directional derivatives verified a posteriori. While case_99 validates this proxy-driven approach, driving optimization natively on side-force derivatives would further refine the Pareto frontier.

4.1.3 Rigid-Body Assumption and Aeroelastic Neglect

The computational framework treats all parametric geometries as rigid bodies. However, optimized designs push toward slender, high-aspect-ratio planforms (AR ≈ 3.0) on a thin 7% SD8020 profile. Under high sideslip angles or dynamic gust encounters, these slender structures are susceptible to aeroelastic flutter, chordwise bending, and torsional twist, which are phenomena that went uncaptured by static OpenFOAM fluid domains.

4.2 Future Work

4.2.1 Transient Solvers and Dynamic Derivative Extraction

While steady-state RANS simulations (k-ω SST-LM) accurately resolve static stability derivatives ($C_{Y\beta}$, $C_{n\beta}$) at fixed angles of sideslip, they cannot capture time-dependent aerodynamic phenomena during dynamic maneuvering. To evaluate non-stationary flow behavior, future iterations of this framework will transition from steady-state models to unsteady RANS (URANS) or Scale-Adaptive Simulation (SAS) in OpenFOAM.

By implementing dynamic mesh motion, forced pitching and yawing oscillations can be prescribed according to:

$$\beta(t) = \beta_0 + \beta_A sin(\omega t)$$

where β_0 represents the mean sideslip angle, β_A is the oscillation amplitude, and ω is the reduced frequency.

This transient approach enables the direct extraction of dynamic damping and rotary derivatives, specifically the yaw damping derivative (C_{nr}) and roll-due-to-yaw derivative (C_{lr}). Furthermore, resolving time-accurate pressure fields will quantify phase lag between forced surface motion and transient side-force generation caused by dynamic laminar separation bubble (LSB) migration. Incorporating these dynamic derivatives will ultimately furnish full 6-DOF aerodynamic transfer functions for micro-UAV closed-loop flight control and autopilot tuning.